PoE (primary side, isolated 1500 Vrms) -> isolated forward converter -> VBUS 48 V -> rails and GaN bridges

POWER AND AMPLIFIER

poe_in
PoE++ input: magjack, bridges, PD
(block A)
File: poe_in.kicad_sch

iso_dcdc
Isolated active-clamp forward, 48 V
(block B)
File: iso_dcdc.kicad_sch

rails
Secondary rails and sequencing
(block C)
File: rails.kicad_sch

amp_woofer
GaN BTL power stage: woofer
(block D)
File: amp_woofer.kicad_sch

iso_aux
Converter sync clock and VBUS droop
(block B)
File: iso_aux.kicad_sch

amp_sense
VBUS, current and temperature sensing
(block D)
File: amp_sense.kicad_sch

amp_tweeter
GaN BTL power stage: tweeter
(block D)
File: amp_tweeter.kicad_sch

VPRI 42-57 V
UCC_SS
VBUS 48 V
VBUS, VDRV 7 V
VBUS, VDRV
ACF_SYNC, DRV_FWD
I, V, T sense
ETH pairs, LEDs
PD status (ISO7740)
POE_LOSS_N, ACF_PULSE
PWM, FAULT, ARM (both bridges)

COMPUTE, NETWORK, AUDIO AND I/O

ethernet
Ethernet PHY and clocks
(block E)
File: ethernet.kicad_sch

fpga
ECP5 FPGA and configuration
(block E)
File: fpga.kicad_sch

mcu
RP2350B, flash, USB
(block E)
File: mcu.kicad_sch

audio_out
1-bit DAC, headphone and line out
(block F)
File: audio_out.kicad_sch

io
User I/O, debug and test points
(block G)
File: io.kicad_sch

audio_in
Measurement mic and MEMS mic
(block F)
File: audio_in.kicad_sch

RMII + 50 MHz
PIO link, SSPI config
1-bit DAC, MCLK_Q, relays
USB, SWD, I2C, LED, display
I2S mic ADC, PDM

Inter-sheet signals are global labels and rails are power symbols; the lines on this page show the main flows.

Net names: design/parts/*-interfaces.md and E-pinmap.md

Open source, MIT
design/intent.md, design/architecture.md
PoE++ GaN speaker amplifier
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /
File: satellite.kicad_sch

Title: Studio Amp satellite, variant A

Size: A3	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 1/14

PoE++ input, block A (references 100–199). VPRI = bridge output = TPS2373 VDD; VPD_N = bridge OUTN = TPS2373 VSS; VPRI_RTN = TPS2373 RTN (after the hot-swap) = block B's primary return.

CBULK (61 uF) is drawn on iso_dcdc: block B's C215 47 uF 100 V Al + C211–C213 3 x 4.7 uF at T1 (>= 10 uF for U101, <= 120 uF; inrush 12.7 ms).

VPRI

VPRI, VPRI_RTN, VB_PRI: block B (iso_dcdc)

PAIR SET A (pins 1–2 / 3–6)

PAIR SET B (pins 4–5 / 7–8)

PWR_FLAG

PWR_FLAG

Q101 PSMN040

Q102 PSMN040

Q103 PSMN040

Q104 PSMN040

Q105 PSMN040

Q106 PSMN040

Q107 PSMN040

Q108 PSMN040

TG12

BG12

TG36

BG36

TG45

BG45

TG78

BG78

POE_CT12

POE_CT36

POE_CT45

POE_CT78

VPD_N

Two ideal-diode bridges. High-side FET: drain VPRI, source CT; low-side: drain CT, source VPD_N.
PSMN040-100MSE: 25–57 mV drop at 0.865 A per pair set, 0.20 W for all eight hot (A-bridge-lt4321.md).
Layout: each pair set's 4 FETs round its two CT pins; TG/BG traces short to U101.

J101 LPJK6078AHNL

4x 1CT:1CT, Bob Smith inside

LINE SIDE (primary)

PHY SIDE (secondary)

CT1–2

CT3–6

CT4–5

CT7–8

MD3+

MD3–

MDCT3

MD4+

MD4–

MDCT4

LEDY+

LEDG+

LEDY–

LEDG–

SH

MD1+

MD1–

MD2+

MD2–

MDCT1

MDCT2

ETH_TXP

ETH_TXN

ETH_RXP

ETH_RXN

ETH_AVDD

ETH_LED_G_A

ETH_LED_Y_K

GND

C101 1n 2kV X7R 1206

R101 2.0M 1% 1206

R102 2.0M 1% 1206

R103 2.0M 1% 1206

R104 2.0M 1% 1206

R105 2.0M 1% 1206

SECONDARY (GND)

Isolation barrier (R-32, 1500 Vrms): this box edge, through J101

SHIELD -> GND: C101 1 nF 2 kV || R101–R105 (10 Mohm).
R-32 hipot: <= 1.2 kVpk on J101's internal 1 nF, <= 1.23 kVpk on C101, AC hipot only.
ESD +-8 kV: 1.33 kV peak, 267 V per resistor.
A-shield-network.md

MD3/MD4 windings unused (100BASE-TX uses pairs 1–2 and 3–6): their line-side CTs still carry power (CT4–5, CT7–8).

U101 LT4321IUF

IN12

IN36

IN45

IN78

OUTP

OUTN

EN_N

EN

R106 100k 1% 0603

R107 1M 1% 0603

Q109 BSS123

D101 BZT52C27

D102 BZT52C10

R108 2.2M 1% 0805

C102 1n 50V X7R

LT4321 enable (A-bridge-lt4321.md calc 3): on for VPRI >= 32.8 V (D101, R106), held off below 21 V (R107). Q109 pulls EN low whenever the TPS2373 hot-swap is open (RTN-VSS > 2.8 V), so the bridges run only with CBULK connected (LT4321 p.7). R108 2.2 M keeps the detection signature at 24.4–24.9 k.

D103 SMCJ58A TVS

D103: <= 93.6 V at 16 A (abs max 100 V). C103 = C_PD, the only cap on VDD–VSS (0.05–0.12 uF). Place both at U101.

C103 100n 100V X7R 0805

R109 24.9k 1% 0603

TPS2373-4 U102

VDD

DEN

CLSA

CLSB

REF

MPS_DUTY

NC

VSS

VC_OUT

VC_IN

PG

TPH

TPL

BT

AMPS_CTL

UVLO_SEL

APD

RTN

PD_PG

C104 1u 50V X7R 0805

C105 100n 50V X7R

VC_OUT unused (1 uF only). VC_IN = VB_PRI, 14.3 V from block B (TPS2373 7.3.7).

R110 63.4R 1% 0805

R111 90.9R 1% 0805

R112 49.9k 1% 0805

R113 1.3M DNP

R114 10k 1% 0603

R115 10k 1% 0603

R116 10k 1% 0603

VISO_P5V

Class B: CLSA 63.4R, CLSB 90.9R (TPS2373 Table 1). RDEN 24.9k (trim to 24.3k if the signature reads high). AMPS_CTL: R113 DNP, auto-MPS off (R-30). MPS_DUTY, UVLO_SEL open: APD to RTN.
A-pd-tps2373.md, A-pd-passives.md

U104 78L05

C107 330n 50V X7R 0805

C108 100n 50V X7R

U104 78L05: SOT-89 O-G-I pinout is the -AB3-R order only.
VISO_P5V (primary): <= 15.3 mA, 177 mW at VB_PRI 14.8 V.

R117 10k 1% 0603

UCC_SS

Q110 2N7002

PD_PG_N LOW = SS/SD above the 2N7002 threshold (1–2.5 V): PD out of inrush and UCC2897A running. UCC_SS is block B's SS/SD node after its 1 k from PD_PG; the gate draws <= 80 nA.

SECONDARY (GND, +3V3)

+3V3

C109 100n 50V X7R

C110 100n 50V X7R

U103 ISO7740DWR

VCC1

INA

INB

INC

IND

NC

GND1

VCC2

OUTA

OUTB

OUTC

OUTD

GND2

EN2

PD_BT_N

PD_TPL_N

PD_TPH_N

PD_PG_N

All four active LOW. ISO7740 (no F suffix) reads HIGH while the primary is unpowered. TPH/TPL/BT valid 20–28 ms after PG releases.
Truth table: A-pd-tps2373.md

Primary = VPRI / VPD_N / VPRI_RTN; secondary = GND. Barrier R-32: 1500 Vrms (dashed).
Block A. Audits: design/parts/A-*.md (values, calculations, datasheet pages)

Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /poe_in/
File: poe_in.kicad_sch

Title: Studio Amp satellite: PoE++ input: magjack, bridges, PD

Size: A2	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 2/14

PRIMARY: VPRI_RTN

ISOLATION BARRIER 1500 Vrms (R-32)

SECONDARY: GND

T1 wired 1:2: primary N1 || N2 = 4 T (Lm 25 uH). secondary N3 + N4 in series (10-8) = 8 T. Dots = winding starts (B-transformer.md). Layout: N1 and N2 symmetric.

N5 aux, 4 T: forward pulse - VPRI, averaged -> LINEOV: output OVP at VBUS = 53.3 V, independent of the opto (B-controller.md)

PRIMARY INPUT CAPACITORS

VPRI

C215 47u 100V 105C

C211 4.7u 100V X7S 1210

C212 4.7u 100V X7S 1210

C213 4.7u 100V X7S 1210

C214 100n 100V X7R 0603

VPRI_RTN

VPRI from the PD (block A), 38.4-57 V, <= 1.85 A. C215 47u: LT4321 needs >= 10 uF; damps the cable. Layout: C211-C214 at T1 pins 2/3 and Q201's Rcs, the 737 kHz pulse-current loop (1.34 A rms). B-primary-power.md

OUTPUT FILTER, IBUS SHUNT 30 mR, VBUS BULK

L202 68u: 0.35-0.55 A p-p, 1.65 A peak at CC. C231-C236 6 x 4.7u 100V ahead of the shunt keep 737 kHz ripple out of it. B-output.md

VBUS 47.9 V, <= 1.30 A from the converter (CC, 62 W): bursts above that come from the bulk

L202 68u CDRH127/LDNP

VBUS_CONV

C231 4.7u 100V X7S 1210

C232 4.7u 100V X7S 1210

C233 4.7u 100V X7S 1210

C234 4.7u 100V X7S 1210

C235 4.7u 100V X7S 1210

C236 4.7u 100V X7S 1210

GND

R233 30m 1% 2512 2W

VBUS

C241 1000u 63V KZN 105C

C242 1000u 63V KZN 105C

C243 1000u 63V KZN 105C

C244 1000u 63V KZN 105C

C245 1000u 63V KZN 105C

C246 1000u 63V KZN 105C

C247 1000u 63V KZN 105C DNP

PWR_FLAG

D208 SMCJ51A

VBUS 47.9 V +-1.5 %: 6 x 1000u 63V KZN (C247 DNP footprint). OVP: SMCJ51A + LINEOV 53.3 V + gateware 55 V (agreed with block D)

T201 750343576 (1:2)

LINEOV

R209 470k 1% 0805

R208 47k 1% 0805

D202 BAV21W 250V

VPRI_RTN

N5 4T

N1 || N2 = 4 T: dots 2+3 -> VPRI 4+5 -> SW_PRI

N1 4T

N3 4T

N2 4T

N4 4T

SW_SEC

UCC2897A CONTROLLER, PRIMARY STAGE

VIN: 100k from VPRI (JFET start-up unused; VDD from VB_PRI). R219 DNP option.

VB_PRI

PWR_FLAG

R220 0R 0603

VREF_ACF

C202 1u 25V X7R 0603

C203 10u 25V X7R 1206

C204 47n 25V X7R 0603

VPRI_RTN

VB_PRI

VPRI

R219 0R 0603 DNP

R218 100k 1% 0805

R206 73.2k 1% 0805

R207 2.37k 1% 0603

U201 UCC2897APWR

VIN, LINEUV, SYNC, FB, LINEOV, SS/SD, RTDEL, RON, ROFF, RSLOPE, GND, PGND, VDD, PVDD, VREF, OUT, AUX, CS

ACF_FB

ACF_SYNC

LINEOV

C207 47n 25V X7R 0603

R210 23.7k 1% 0603

LINEUV: on 40.4 V, off 38.4 V

UCC_SS

C205 1u 25V X7R 0603

R203 4.53k 1% 0603

R201 31.6k 1% 0603

R202 14.3k 1% 0603

R204 316k 1% 0603

RDEL 4.53k = 50 ns. RON 31.6k / ROFF 14.3k: free-run 656 kHz (590-708), synced to 737.28 kHz; DMAX 0.68-0.715. RSLOPE 316k. CSS 1u: 60 ms soft start. PD_PG holds SS low during PD inrush; UCC_SS -> block A. GND (8) and PGND (13) meet only at the IC (datasheet 11.1). B-controller.md

R217 1k 1% 0603

PD_PG

LINEOV: R209 470k from node A: 23.7k || 47n averages (1 ms)

R212 10k 1% 0603

R213 2.2R 1% 0603

R214 10R 1% 0603

R216 68m 1% 2512 2W

CS_PRI

C208 82p 50V C0G 0603

R205 1k 1% 0603

C209 100n 250V X7R 1206

C210 100n 250V X7R 1206

Q201 FDMS86200 150V

SW_PRI

Q202 IRFR6215 -150V

R215 10k 1% 0603

D201 BAS316

VPRI_RTN

Active clamp. P-FET low side: Vc ~ 100 V on C209 || C210 (250 V X7R). B-primary-power.md

C223 2.2n Y2/X1 250VAC 2220

C224 2.2n Y2/X1 250VAC 2220 DNP

C223 Y2 2.2n at T1: interwinding displacement current. C224 DNP (EMI tuning). B-feedback.md

FORWARD SR

SR_FWD_D

C225 1u 25V X7R 0603

U204 NCP4306DADZZDA

MIN_TON, MIN_TOFF, VCC, CS, DRV, GND

R227 750 1% 0603

R226 2k 1% 0603

C227 22p 250V C0G 0805

R231 47R 2512 1W

D203 US1G 400V

C229 10n 250V X7R 1206

SMBJ130A

Q203 BSC500N20NS3 200V

DRV_FWD

FREEWHEEL SR

C226 1u 25V X7R 0603

U205 NCP4306DADZZDA

R228 750 1% 0603

R229 2k 1% 0603

C228 22p 250V C0G 0805

R232 47R 2512 1W

D204 US1G 400V

C230 10n 250V X7R 1206

SMBJ130A

Q204 BSC500N20NS3 200V

Both SRs: RC snubber 22p + 47R 2512; RCD-Z clamp (US1G, 10n, SMBJ130A) holds the drain and NCP4306 CS <= -165 V (200 V abs max). MIN_TON 750R - 125 ns, MIN_TOFF 2k - 256 ns. Layout: NCP4306 GND Kelvin to its SR source pad. CS to the drain pad. B-sync-rect.md

IBUS SENSE (NA281)

VSR

C208 100n 50V X7R 0603

U206 INA281A4

Vin+, Vin-, V+, OUT, GND

R234 100 1% 0603

C237 470n 16V X7R 0603

R235 200k 1% 0603

R236 100k 1% 0603

IBUS_F

IBUS_SENSE

INA281A4 200 V/V x 30 mR = 6.0 V/A: 100R + 470n kills 737 kHz ripple. IBUS_SENSE 2.0 V/A, 67k source -> RP2350 ADC4 (block E fits 100 nF). B-feedback.md

OPTO LED, CV LOOP (TL431) AND CC LOOP (TL431, ILIM_SEL)

VSR

R245 499k 1% 0603

R246 115k 1% 0603

C240 4.7n 50V X7R 0603

VBUS

R244 47.5k 0.1% 0603

R248 2.61k 0.1% 0603

U203 LTV-217-B

R211 4.99k 1% 0603

C206 6.8n 50V X7R 0603

OPTO_K

D207 BAS316

U208 TL431B

FB: 4.99k to VREF, 6.8n (pole 4.7 kHz). Opto gain CTR x 4.99k / 499R = 10. fc = 1 kHz, PM = 66 deg (B-feedback.md)

CV: 2.495 V x (1 + 47.5k/2.61k) = 47.9 V. Type II: 115k + 4.7n (zero 294 Hz).

R243 10n 50V X7R 0603

C239 100k 1% 0603

IBUS_F

R237 4.99k 1% 0603

R238 88.7k 1% 0603

U207 TL431B

R239 4.22k 1% 0603

R240 5.9k 1% 0603

ILIM_SEL0

ILIM_SEL1

R241 100k 1% 0603

R242 100k 1% 0603

Q205 AO3400A

Q206 AO3400A

CC limit (ILIM_SEL1:0, reset 00 via 100k): 00 0.44 A, 01 0.94 A, 10 0.80 A, 11 1.30 A. Integrator 100k + 10n. B-feedback.md

VB_PRI BIAS BUCK (PRIMARY)

VPRI

C222 100n 100V X7R 0603

C221 1u 100V X7R 0805

R221 267k 1% 0805

U202 LM5019

VIN, RON, ULVO, BST, SW, FB, VCC, RTN, RTNPAD

C219 10n 50V X7R 0603

C218 1u 25V X7R 0603

L201 330u 0.34A

C216 1n 50V C0G 0603

R226 200k 1% 0603

C217 100n 50V X7R 0603

R222 107k 1% 0603

R223 10k 1% 0603

C220 10u 25V X7R 1206

VB_PRI

R224 237k 1% 0805

R225 10k 1% 0603

VB_PRI 14.3 V (13.9-14.8), <= 85 mA: UCC2897A, block A's TPS2373 VC_IN and isolator, the sync clock (iso_aux). Starts at VPRI 30.3 V. 600 kHz (RON 267k): Type III ripple Rr 200k, Cr 1n, Cac 100n. B-bias.md

Active-clamp forward, 737.28 kHz synced, T1 1:2, VPRI 38.4-57 V -> VBUS 47.9 V
Block B. Audits: design/parts/B-*.md (overview B-converter-design.md)
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /iso_dcdc/
File: iso_dcdc.kicad_sch

Title: Studio Amp satellite: Isolated active-clamp forward, 48 V

Size: A2	Date: 2026-11-08	Rev: A
KiCad E.D.A. 9.0.8		Id: 3/14

ISOLATION BARRIER

PRIMARY SYNC CLOCK (VPRI_RTN) -> UCC2897A SYNC

U252 HT7550-1 5V (VIN, GND, VOUT) — VB_PRI, V5_CLK, VPRI_RTN

V5_CLK 5 V, <= 10 mA: oscillator, counter, one-shot (0.1 W in the SOT-89)

U253 74LVC1GU04 — 7.3728 MHz, CL 20 pF, +-50 ppm all-in (SXMH7.3728F2020F30, -40..85 C): 33p + 33p + ~5 pF stray = 21.5 pF. Rf 1M, Rs 1k (drive ~70 uW).

Y251 7.3728MHz 20pF; C260, C261 33p 50V C0G 0603; R257 1M 1% 0603; R258 1k 1% 0603

U254 74HC4017 — CLK, CKEN, Reset, VDD, VSS, Q0..Q9, Cout

Only this gate runs from VREF, so SYNC <= VREF + 0.3 V in every state; 10k holds SYNC low when VREF is off (IOFF output). VREF budget <= 0.6 mA of 5 mA. B-sync-clock.md

U255A 74LVC1G123 — R259 10k 1% 0603; C264 470p 50V C0G 0603

U255B 74LVC1G123 — C265 100n 50V X7R 0603

U256 74LVC1G08 — R260 100 1% 0603; R261 10k 1% 0603; C266 100n 50V X7R 0603; VREF_ACF; ACF_SYNC

Q1: one 136 ns pulse per 10 clocks = 737.28 kHz (< (1 - DMAX) x T = 0.39 us). CP1 and MR low.

Clock-OK: retriggered by CO; Rext 10k / Cext 470p -> tw ~ 4.5 us (> 1.36 us CO period, <= 10 us). Clock stops -> CLK_OK low within ~4.5 us -> SYNC low -> UCC2897A free-runs (590-708 kHz). B triggers on A falling edges with B held high.

Nothing crosses here: the clock stays primary, the detector and buffer are secondary.

VSR 10 V BIAS BUCK (SECONDARY)

U251 LM5019 (VIN, RON, ULVO, BST, SW, FB, VCC, RTN, RTNPAD) — L251 220u 0.36A

C252 100n 100V X7R 0805; C251 1u 100V X7R 0805; R251 280k 1% 0805; R254 51.1k 1% 0805; R255 10k 1% 0603; R256 82k 1% 0603; C256 1n 50V C0G 0603; C254 10n 50V X7R 0603; C257 100n 50V X7R 0603; C253 1u 25V X7R 0603; R252 71.5k 1% 0603; R253 10k 1% 0603; C255 10u 25V X7R 1206; VBUS; VSR; PWR_FLAG

VSR 10 V (9.7-10.3), <= 37 mA: NCP4306s, TL431s, opto LED, INA2B1, droop comparator. Starts at VBUS 7.48 V (51.1k/10k), dropout below ~10.6 V. 400 kHz (RON 280k); Type III Rr 82k, Cr 1n, Cac 100n. B-bias.md

ACF_PULSE BUFFER (+3V3)

U259B 74LVC2G08 — R270 10k 1% 0603; R271 4.99k 1% 0603; DRV_FWD; ACF_PULSE

U259A 74LVC2G08

U259C 74LVC2G08 — C269 100n 50V X7R 0603

DRV_FWD (10 V forward-SR gate) / 3 -> 3.33 V -> gate B (+3V3, 5.5 V-tolerant inputs) -> ACF_PULSE: one edge per conduction pulse, 737.28 kHz locked. FPGA B6 via block E's 4.7k; clock-fault rule in B-interfaces.md. Gate A unused.

VBUS DROOP DETECTOR

U258 TL431B — R262 3.9k 1% 0603; R263 150k 0.1% 0805; R264 10.0k 0.1% 0603; C267 33n 50V X7R 0603; VBD

U257A LM393 — R265 10k 1% 0603; R266 866k 1% 0603; R267 3.3k 1% 0603; R268 10k 1% 0603; R269 15k 1% 0603; Q251 2N7002; POE_LOSS_N

U257C LM393 — C268 100n 50V X7R 0603

U257B LM393 — VREF_D

POE_LOSS_N low when VBUS < 39.5 V (38.5-40.5), released > 41.1 V (40.0-42.2): 150k/10k 0.1 %, 33n: tau 0.31 ms. 2N7002 via 10k/15k (gate 0.42 V max when off). Pull-up 4.7k to +3V3 at the FPGA (block E). B-vbus-droop.md

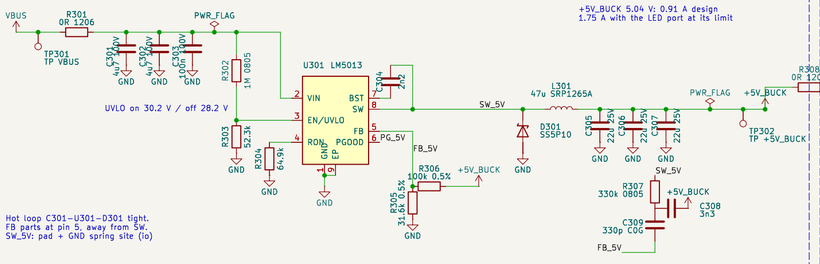
737.28 kHz sync clock, VSR bias, VBUS-droop detector, ACF_PULSE
Block B. Audits: design/parts/B-sync-clock, B-bias, B-vbus-droop, B-interfaces
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /iso_aux/
File: iso_aux.kicad_sch

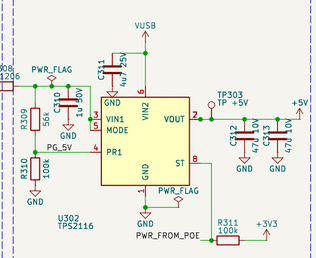
Title: Studio Amp satellite: Converter sync clock and VBUS droop

Size: A3	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 4/14

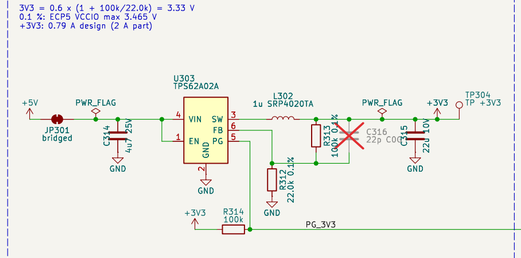
5 V BUCK LM5013 ~213 kHz (C-5v-buck.md)



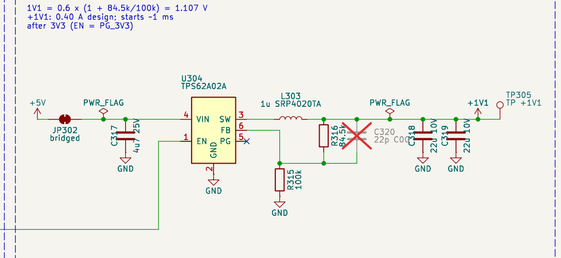
5 V SOURCE SELECT TPS2116
buck has priority (C-5v-mux.md)



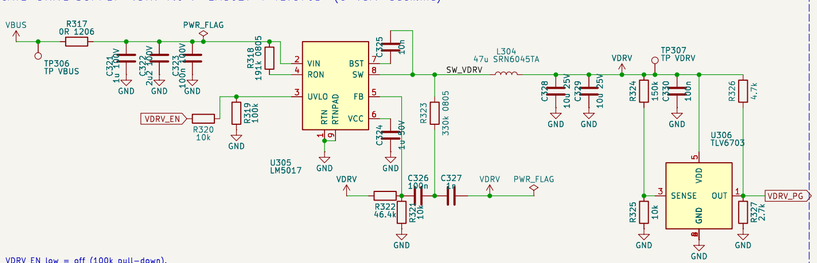
3V3 BUCK TPS62A02A 2.4 MHz FPWM



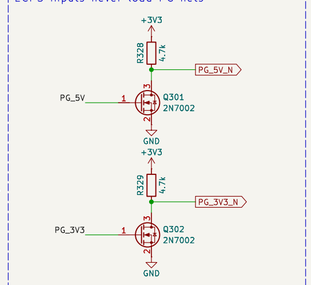
1V1 BUCK TPS62A02A (ECP5 VCC)



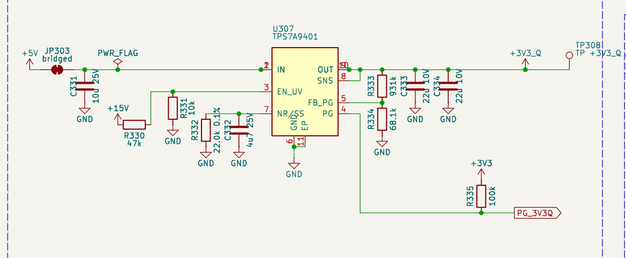
GATE-DRIVE SUPPLY VDRV 7.0 V LM5017 + TLV6703 (C-vdrv-buck.md)



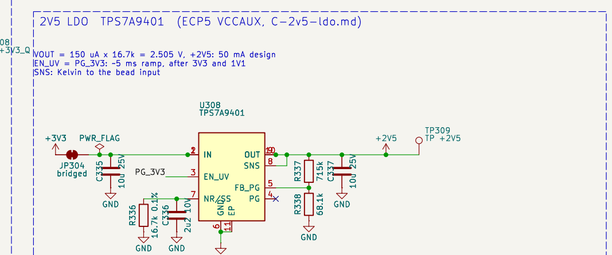
POWER-GOOD BUFFERS
ECP5 inputs never load PG nets



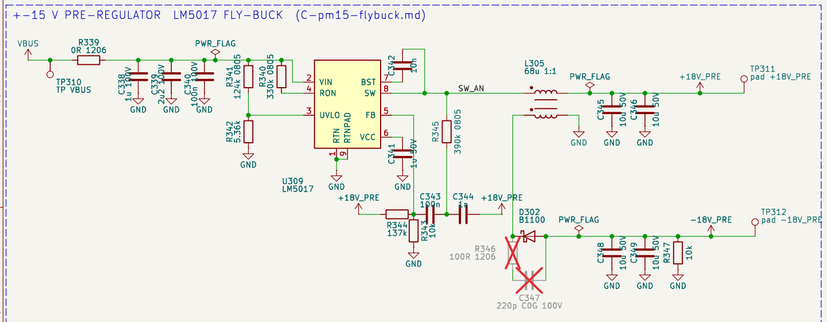
3V3_Q LDO TPS7A9401 (DAC reclock, C-3v3q-ldo.md)



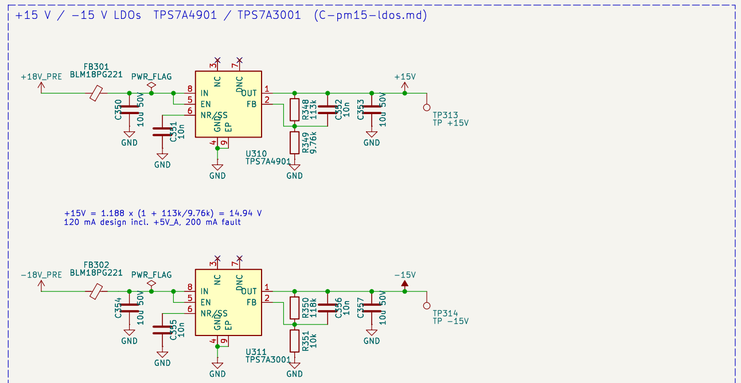
2V5 LDO TPS7A9401 (ECP5 VCCAUX, C-2v5-ldo.md)



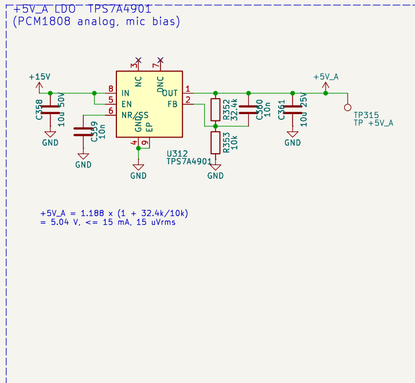
+-15 V PRE-REGULATOR LM5017 FLY-BUCK (C-pm15-flybuck.md)



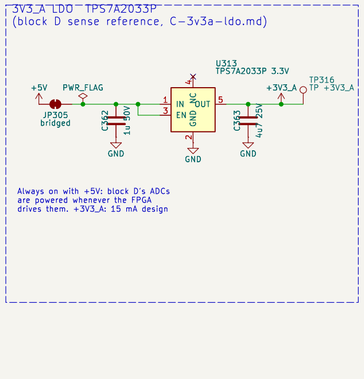
+15 V / –15 V LDOs TPS7A4901 / TPS7A3001 (C-pm15-ldos.md)



+5V_A LDO TPS7A4901
(PCM1808 analog, mic bias)



3V3_A LDO TPS7A2033P
(block D sense reference, C-3v3a-ldo.md)



SEQUENCING (C-sequencing.md)

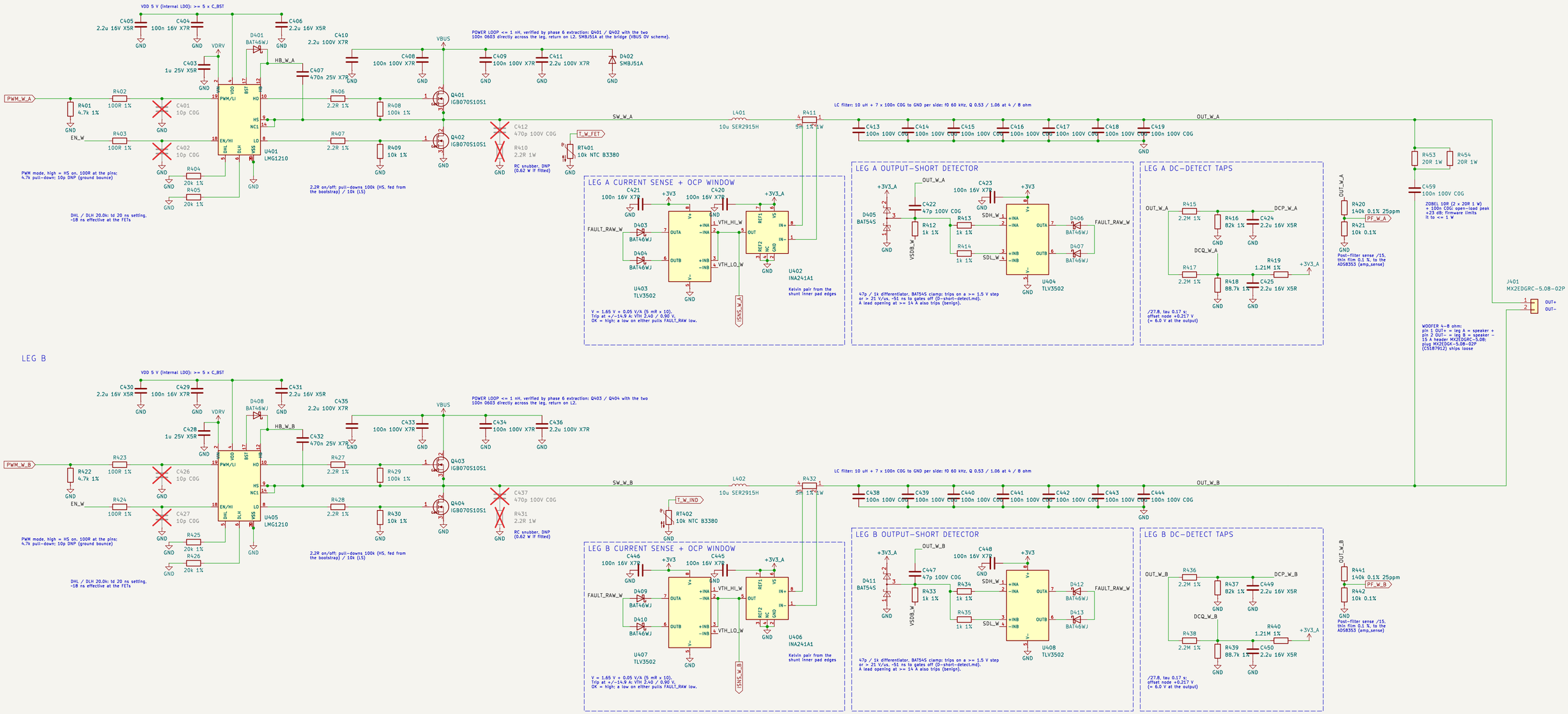
VBUS >= 30.2 V: LM5013 –> +5V_BUCK –> PG_5V
  –> TPS2116 selects the buck (else VUSB)
+5V –> 3V3 –> PG_3V3 –> 1V1 (~1 ms), 2V5 (~5 ms)
VBUS >= 29.8 V: Fly-Buck –> +-15 V (14 ms)
  –> +5V_A: +15V >= 7.1 V enables 3V3_Q
VDRV off until VDRV_EN (firmware, after
  FPGA DONE, PD_PG_N low, POE_LOSS_N high)
USB only: +5V, 3V3, 1V1, 2V5, 3V3_A

Hold-up >= 34 ms after the VBUS < 40 V
trip at 60 W on 6 mF (UVLO-off <= 30.2 V)

R-80: 0R / bridged-jumper link at every
converter input; test point on every rail;
SW-node pads with GND springs on lo sheet

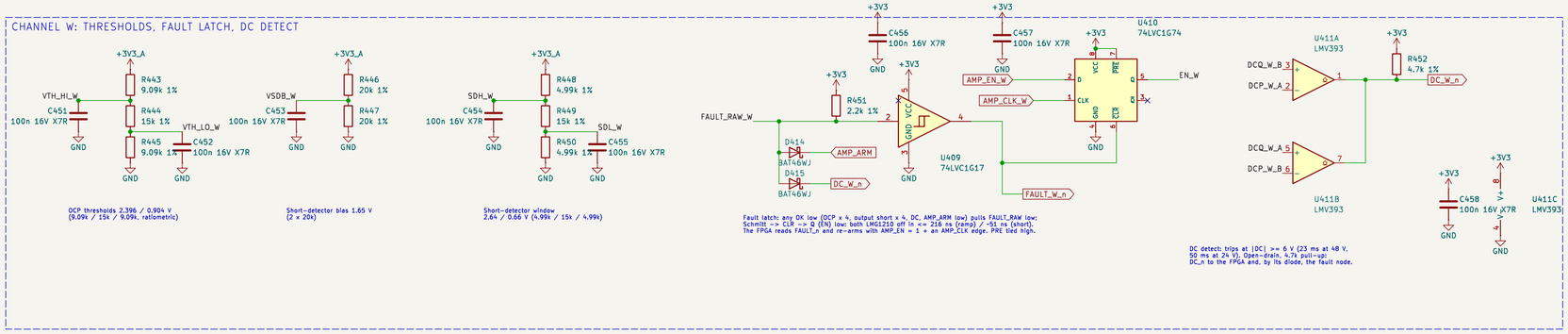
Block C. Audits: design/parts/C-*.md
Sequoia Alexander, github.com/sequoia-hope/studio-amp
Sheet: /rails/
File: rails.kicad_sch
Title: Studio Amp satellite: Secondary rails and sequencing
Size: A2 | Date: 2026-11-08 | **Rev: A**
KiCad E.D.A. 9.0.8 | Id: 5/14

LEG A



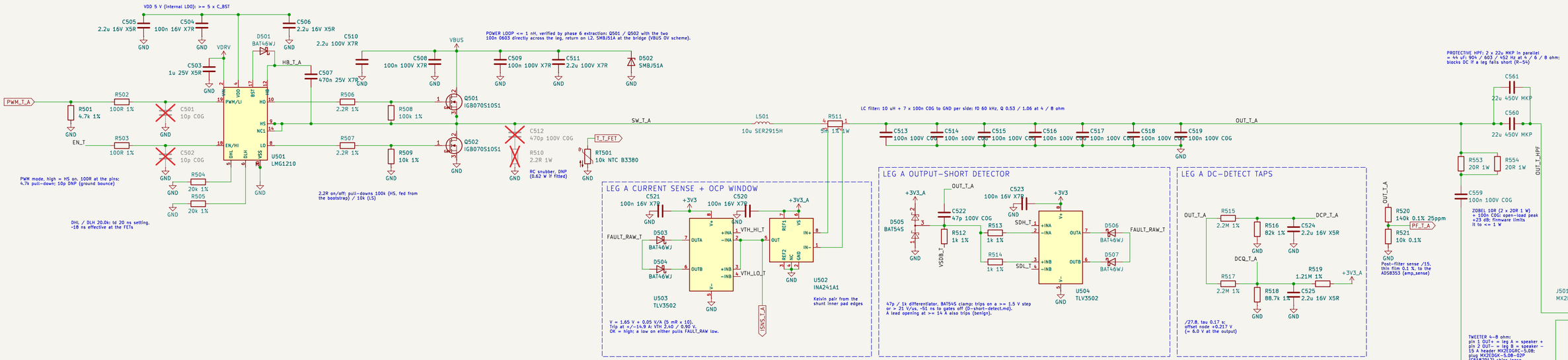
LEG B

CHANNEL W: THRESHOLDS, FAULT LATCH, DC DETECT



gen/amp.py, both channels; the tweeter adds the 2 x 22 uF PP HPF caps
Block D. Audits: design/parts/D-*.md (IGB070S10S1, fsw 576 kHz, D-interfaces.md)
Sequoia Alexander, github.com/sequoia-hope/studio-amp
Sheet: /amp_woofer/
File: amp_woofer.kicad_sch
Title: Studio Amp satellite: GaN BTL power stage: woofer
Size: A2 | Date: 2026-11-08 | **Rev: A**
KiCad E.D.A. 9.0.8 | Id: 6/14

LEG A



LEG B

CHANNEL T: THRESHOLDS, FAULT LATCH, DC DETECT

gen/amp.py, both channels; the tweeter adds the 2 x 22 uF PP HPF caps
Block D. Audits: design/parts/D-*.md (IGB070S10S1, fsw 576 kHz, D-interfaces.md)

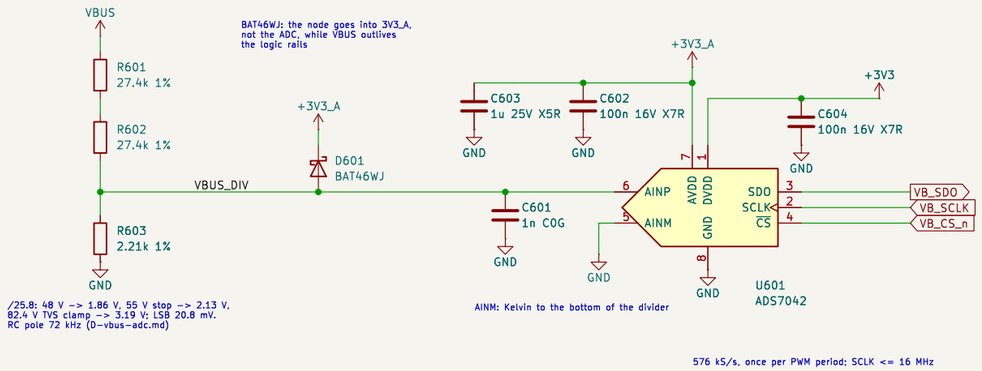
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /amp_tweeter/
File: amp_tweeter.kicad_sch

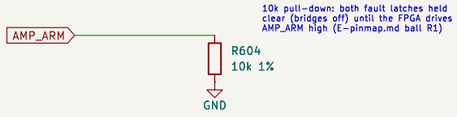
Title: Studio Amp satellite: GaN BTL power stage: tweeter

Size: A2	Date: 2026-11-08	Rev: A
KiCad E.D.A. 9.0.8		Id: 7/14

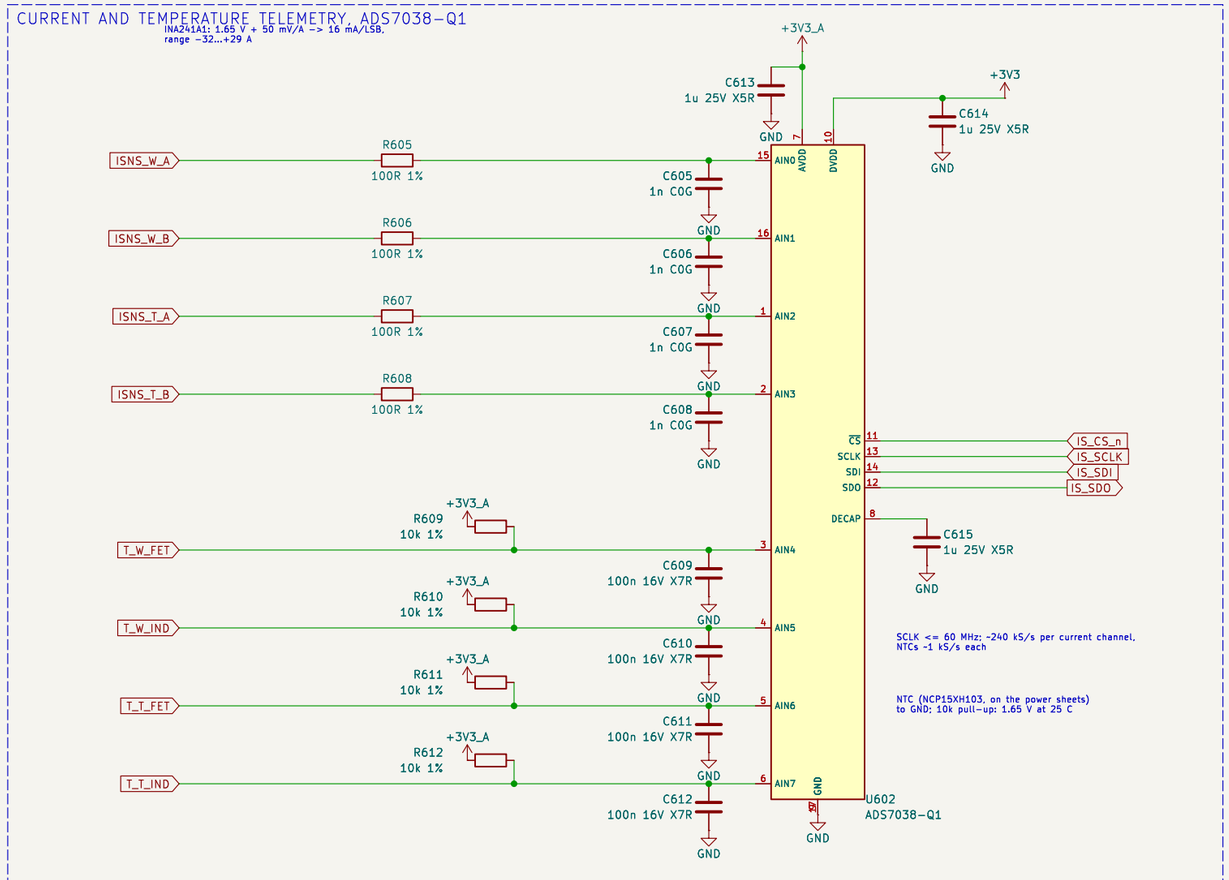
VBUS SENSE, ADS7042 (FEED-FORWARD, POE-LOSS RULE)



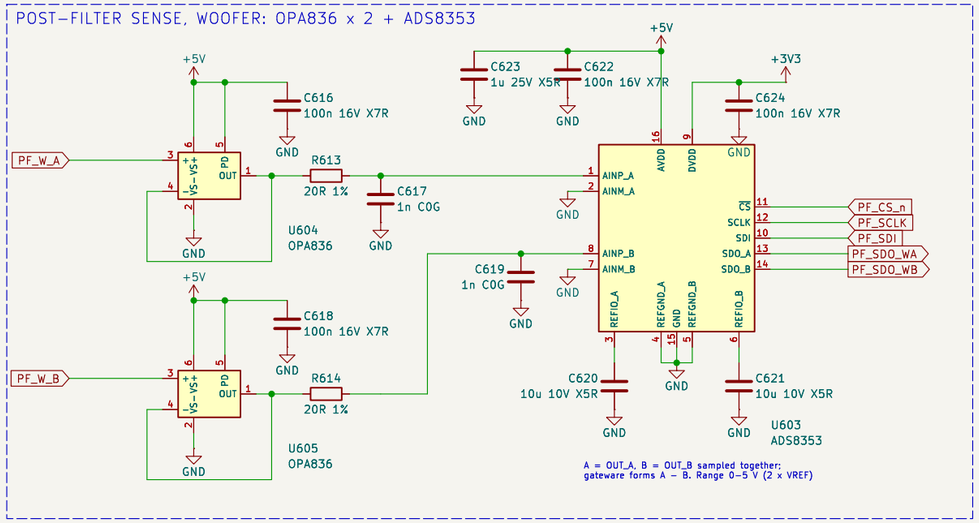
AMP_ARM DEFAULT



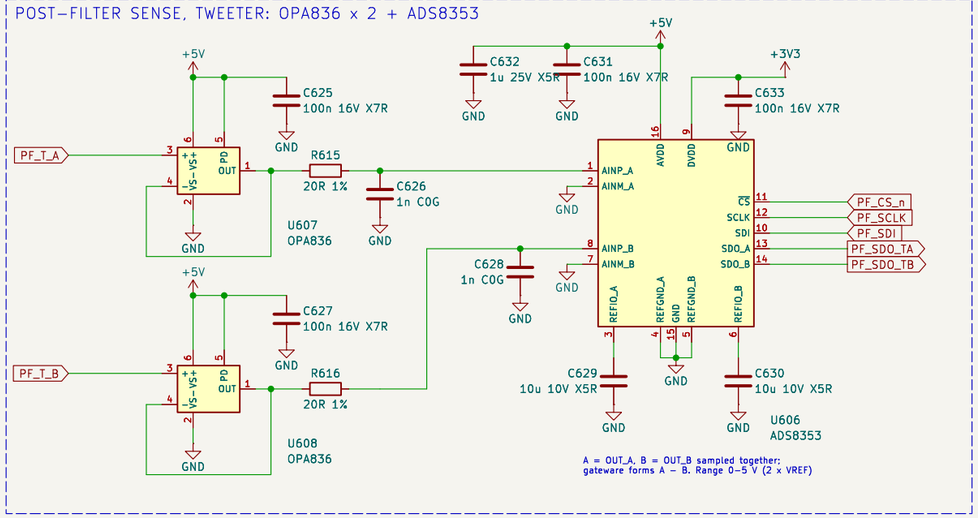
CURRENT AND TEMPERATURE TELEMETRY, ADS7038-Q1



POST-FILTER SENSE, WOOFER: OPA836 x 2 + ADS8353



POST-FILTER SENSE, TWEETER: OPA836 x 2 + ADS8353



PF dividers (140k/10k 0.1%) sit on the power sheets; shared CS/SCLK/SDI, one SDO per ADC channel (D-postfilter-sense.md). +5V load ~24 mA.

+3V3_A (block C LDO): ADC and INA241 reference
Block D. Audits: D-vbus-adc.md, D-isense-adc.md, D-ntc.md, D-postfilter-sense.md

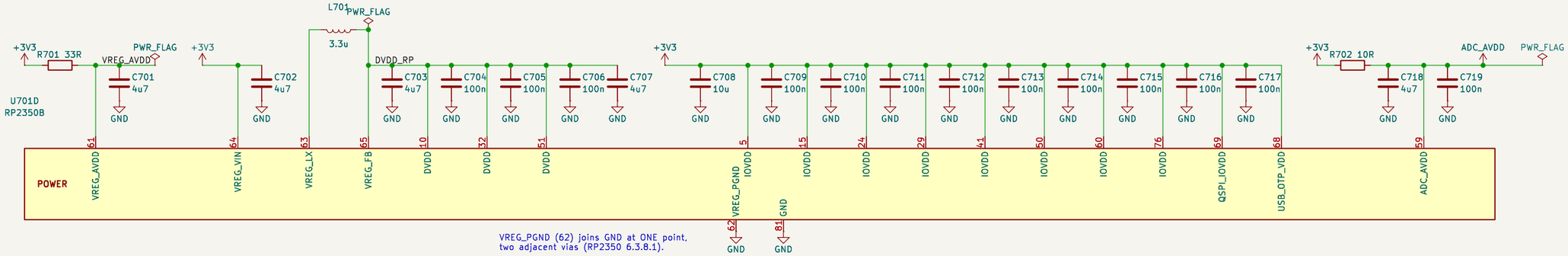
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /amp_sense/
File: amp_sense.kicad_sch

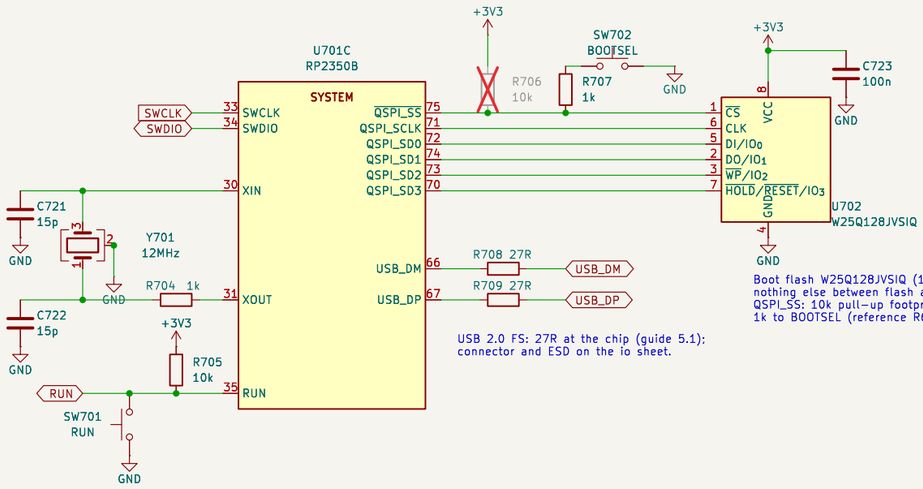
Title: Studio Amp satellite: VBUS, current and temperature sensing

Size: A3	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 8/14

Core regulator, RP2350 6.3.8 / Minimal board: CIN, L and COUT on the chip's side; copy the reference corner (chips/rp2350.md). L: AOTA-B201610S3R3 3.3u, dot to DVDD_RP, reference rotation (orientation changes Vout). DVDD_RP is the RP2350's own 1.1 V (internal regulator); it is not the FPGA's +1V1.



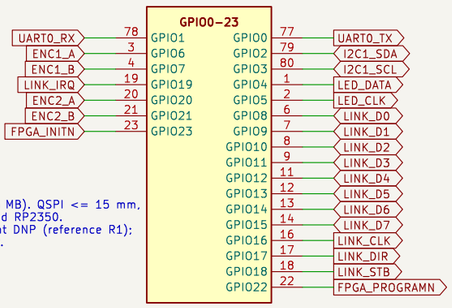
VREG_PGND (62) joins GND at ONE point, two adjacent vias (RP2350 6.3.8.1).



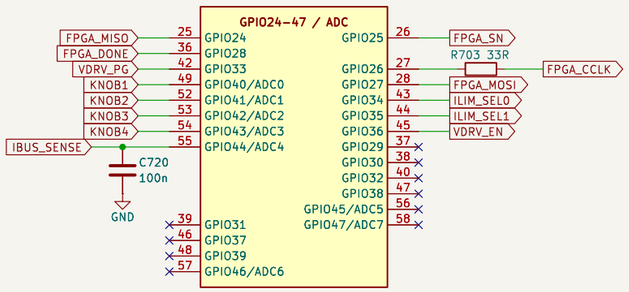
ABM8-272-T3 12 MHz with 15p + 15p and 1k at XOUT: Raspberry Pi's tuned network (guide 4); traces < 5 mm.

USB 2.0 FS: 27R at the chip (guide 5.1); connector and ESD on the io sheet.

Boot flash W25Q128JVSIQ (16 MB). QSPI <= 15 mm, nothing else between flash and RP2350. QSPI_SS: 10k pull-up footprint DNP (reference R1); 1k to BOOTSEL (reference R6).



PIO link to the ECP5 (E-link.md): 8-bit, 37.5 MHz, 22R series at each driver is in the FPGA design rule; LINK_* tri-stated while the FPGA is configured. ENC_*: encoder header on the io sheet (block G pulls).



SSPI configuration (SPI1): FPGA_CCLK 33R at the RP2350, pad set to 8 mA (1k pull-up at the FPGA). Reads <= 18.75 MHz, bitstream writes 37.5 MHz. IBUS_SENSE: 100n at the pin (B's 67k source, 24 Hz). ILIM_SEL0/1, VDRV_EN: reset Hi-Z, pulled down at blocks B and C (safe state).

Regulator, decoupling, crystal and flash copied from the Raspberry Pi RP2350B Minimal board
Block E. Audits: design/parts/E-mcu-rp2350b.md, E-mcu-passives.md, E-xtal-12mhz.md, E-flash-

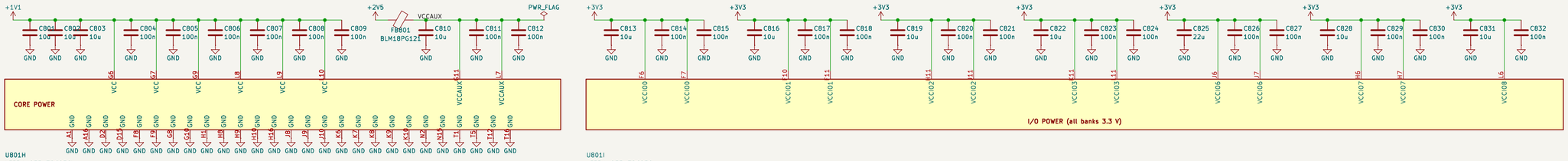
Sequoia Alexander, github.com/sequoia-hope/studio-amp

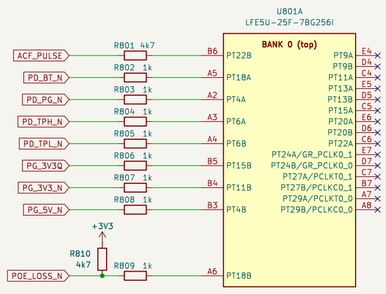
Sheet: /mcu/
File: mcu.kicad_sch

Title: Studio Amp satellite: RP2350B, flash, USB

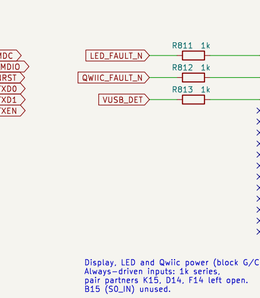
Size: A3	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 9/14

CORE: VCC +1V1 (1.045–1.155 V). 3 x 10u + 100n per ball; VCCAUX +2V5 through a 120 R bead, 10u + 100n per ball (TN-02038 Tab 3.1).
I/O: every VCCIO on +3V3 (max 3.465 V). 10u per bank (22u on bank 6, the PWM bank) + 100n per ball. Layout: 100n under the ball field between the dog-bone vias; bulk just outside the package.

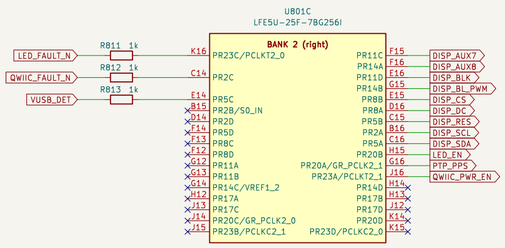




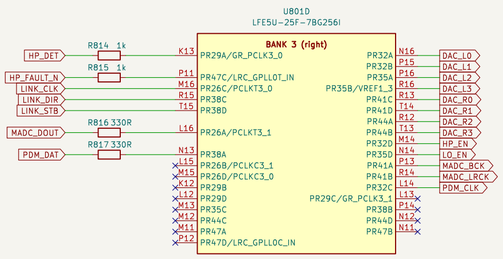
Status from always-powered parts (A, B, C):
1k in series (hot socket, DS-02012 3.6).
POE_LOSS_N: block B open drain, 4k7 pull-up on the net (pre-config sweep, E-pinmap.md).
ACF_PULSE: 4k7 series (B's 74LVC2G08).



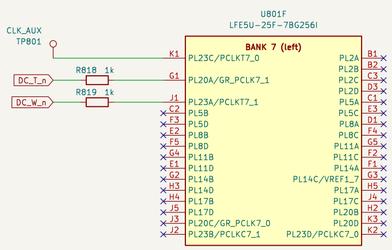
RMII to the LAN8720A (ethernet sheet): REFCLK on PCLKT1_0 (C8), MODE straps (RXD0/1, CRS_DV), MDIO, nRST and nINT pulls sit at the PHY.
LPF: PULLMODE=UP on A11, A12, B11.



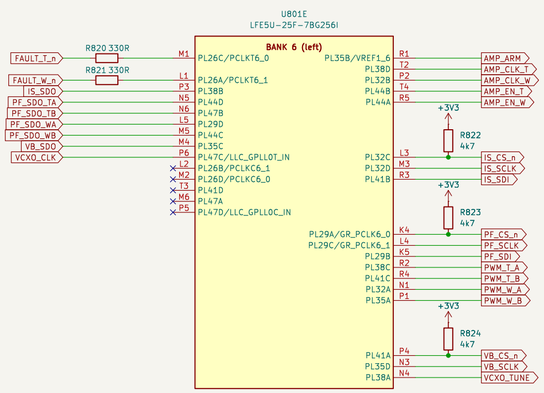
Display, LED and Qwiic power (block G/C).
Always-driven inputs: 1k series,
pair partners K15, D14, F14 left open.
B15 (SO_IN) unused.



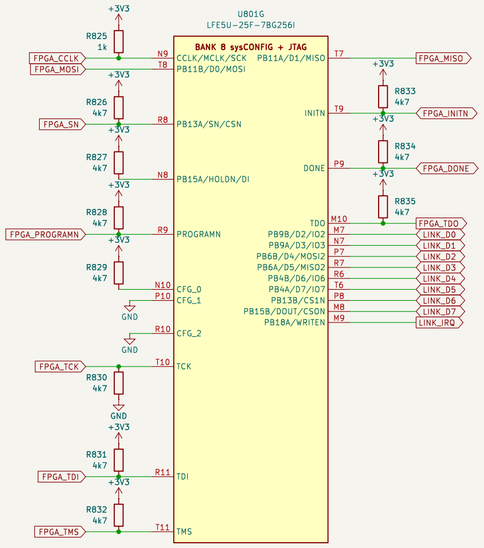
Audio (block F) and PIO link control (RP2350).
MADC_DOUT, PDM_DAT: 330R; HP_FAULT_N, HP_DET: 1k;
partners L15, P14, P12, K12 left open.
LINK_CLK on PCLKT3_0 (M16).



CLK_AUX (K1, PCLKT7_0): spare clock input
on a test pad (external low-jitter clock option).
DC_W_n / DC_T_n: 1k series, partners J2 / H2 open.



Power stage (block D). PWM_* from ODDRX2F at 589.824 Mb/s
(PLL x12 from VCXO_CLK on P6, LLC_GPLL0T_IN: P5 open).
FAULT_*: always driven, 330R, partners L2 / M2 open.
T3 open: it pairs with PWM_T_B (R4).
ADC chip selects: 4k7 pull-ups keep the ADCs
deselected through configuration.



sysCONFIG slave SPI from the RP2350 (SPI1, 37.5 MHz):
CFGMDN[2:0] = 001: CFG0 4k7 up, CFG1/CFG2 to GND.
CCLK 1k pull-up to VCCIO8 (TN-02038 Tab 6.2):
33R series at the RP2350 (mcu sheet).
PROGRAMN, INITN, DONE, SN, HOLDN: 4k7 up.
JTAG: TDI, TMS, TDO 4k7 up, TCK 4k7 down
(TN-02038 Tab 6.1): header on the Io sheet.
LINK_D0–7 / LINK_IRQ: PIO link (E-link.md).

ECP5 LFE5U-25F-7BG256I, slave SPI from the RP2350 (CFGMDN 001). All VCCIO 3.3 V.
Block E. Audits: design/parts/E-fpga-ecp5.md, E-fpga-ecp5-passives.md; nets: E-pinmap.md

Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /fpga/
File: fpga.kicad_sch

Title: Studio Amp satellite: ECP5 FPGA and configuration

Size: A2	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 10/14

AUDIO / PWM MASTER CLOCK, 24.576 MHz VCXO

VCXO_TUNE
R917 100k
R918 100k
R919 0R
C913 1u
C914 1u
R920 0R
VCXO_VDD
C915 100n
C916 4u7
PWR_FLAG
FB902 BLM18PG221
+3V3
Y902 24.576MHz VCXO
VCXO_CLK
R921 33R
MCLK_Q
C904 470p
C903 1u
VDDCR

Tune: 1-bit delta-sigma from ECP5 N4, 2 x (100k, 1u): 1.59 Hz poles,
> 16 bits below 2 Hz; firmware clamps Vc to 0.3-3.0 V.
Fixed-XO fallback on the same land: fit the 0R to VDD, remove the series 0R.

VCXO_CLK -> ECP5 P6 (LLC_GPLL0T_IN), shortest branch, no R.
MCLK_Q -> block F's SN74LVC2G34 (MCLK_Q).
VCXO within 10 mm of P6, RC filter beside Vc,
nothing switching under the 1u caps.

Straps (sampled at nRST release):
MODE[2:0] = 111 (all capable, auto-neg):
4k7 pull-ups beat the ECP5's 150 uA
pre-config pull-down (review MAJOR-2):
gateware: PULLMODE=UP, nRST released
>= 25 ms after the rails.
PHYAD0 = 0 (RXER internal pull-down).

nRST 10k down: PHY held in reset
until the FPGA design releases it.
MDIO 1k5 up (tau 30 ns).

C906 4u7
C905 100n
C907 100n
C908 100n
C909 4u7
FB901 BLM18PG221
C910 100n
C911 100n

CT bypass, place at
J101 pins 12 and 6

ETH_AVDD: analog 3.3 V island (VDD1A, VDD2A, 4 x 49R9,
magnetics centre taps, yellow LED anode) behind a
BLM18PG221 (<= 130 mA, 13 mV). VDDCR: internal 1.2 V
regulator output, 1u || 470p at pin 6 (Tab 2-7).

R901 4k7
R902 4k7
R903 4k7
R904 10k
R905 1k5
R906 10k
R907 10k
RMII_RXD0
RMII_RXD1
RMII_CRSDV
RMII_TXD0
RMII_TXD1
RMII_TXEN
TP901
RXER
PHY_MDC
PHY_MDIO
PHY_NINT
PHY_NRST

U901
LAN8720AI
RXD0/MODE0
RXD1/MODE1
CRS_DV/MODE2
RXER/PHYAD0
TXD0
TXD1
TXEN
MDC
MDIO
INT/REFCLKO
RST
XTAL1/CLKIN
XTAL2
VDDCR
VDDIO
VDD1A
VDD2A
VSS

R908 49R9 1%
R909 49R9 1%
R910 49R9 1%
R911 49R9 1%
TXP
TXN
RXP
RXN
ETH_TXP
ETH_TXN
ETH_RXP
ETH_RXN
C901 100n
C902 100n

49R9 + 100n at the MDI pins, symmetric per pair;
pairs 100 R differential, no vias, over solid GND.
J101 (poe_in): TX -> 11/10, RX -> 4/5.

R912 270R
R913 270R
R914 12k1 1%
LED1/REGOFF
LED2/INTSEL
RBIAS
ETH_LED_G_A
ETH_LED_Y_K

LED1 (link, active high) -> green 20/19:
LED2 (100M, active low) <- yellow 18/17 from VDD2A.
Straps: REGOFF = 0 (internal 1.2 V on).
nINTSEL = 1 (REF_CLK in). RBIAS 12k1 1%.

C912 100n
R915 33R
R916 33R
Y901 50MHz
RMII_REFCLK

RMII REFERENCE CLOCK: 50 MHz +-50 ppm CMOS XO, OE tied high.
Two point-to-point branches, 33R each at the XO:
PHY XTAL1 and ECP5 C8 (PCLKT1_0), <= 25 mm,
matched +-5 mm, over solid GND.

LAN8720AI in REF_CLK-in mode (nINTSEL = 1), MODE 111, PHYAD 0
Block E. Audits: design/parts/E-phy-lan8720.md, E-phy-passives.md, E-clk-rmii-50mhz.md, E-clk-

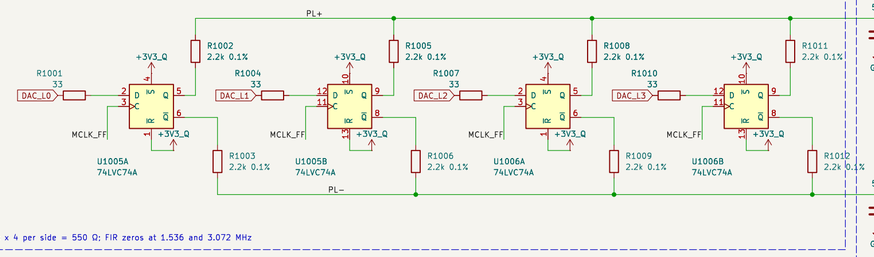
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /ethernet/
File: ethernet.kicad_sch

Title: Studio Amp satellite: Ethernet PHY and clocks

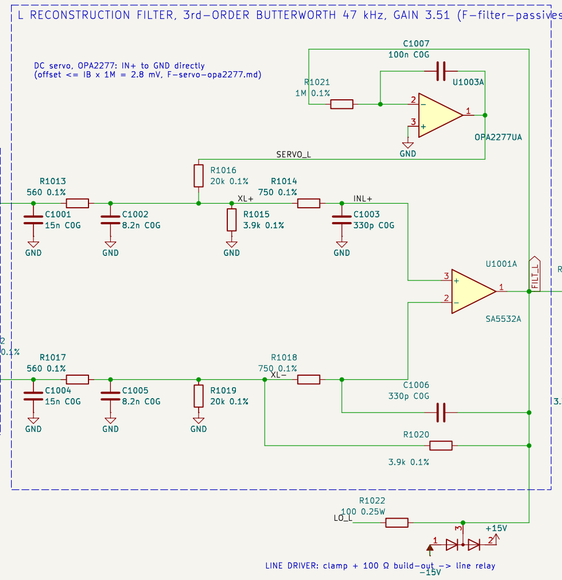
Size: A3	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 11/14

L 1-BIT DAC: 4-TAP FIR, 74LVC74 ON +3V3_Q, CLOCKED BY MCLK_FF (F-dac-74lvc74.md)

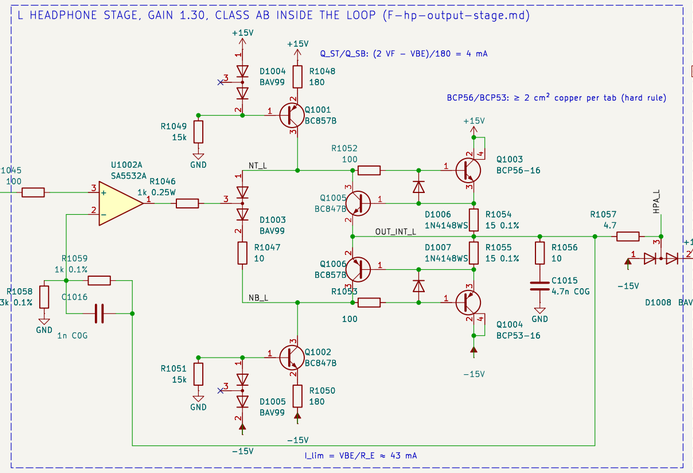


R_T 2.2k x 4 per side = 550 Ω; FIR zeros at 1.536 and 3.072 MHz

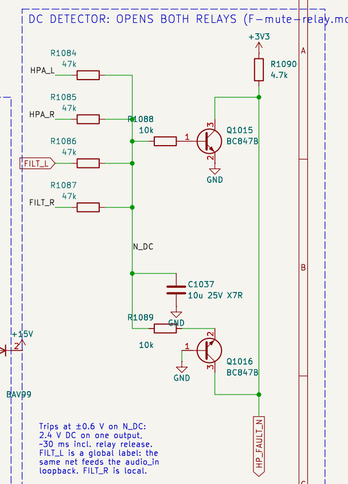
L RECONSTRUCTION FILTER, 3rd-ORDER BUTTERWORTH 47 kHz, GAIN 3.51 (F-filter-passives.md)



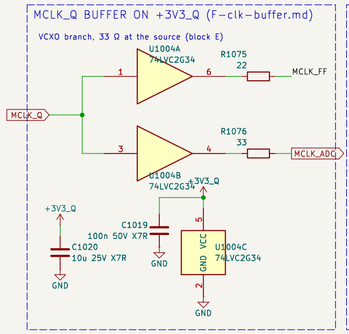
L HEADPHONE STAGE, GAIN 1.30, CLASS AB INSIDE THE LOOP (F-hp-output-stage.md)



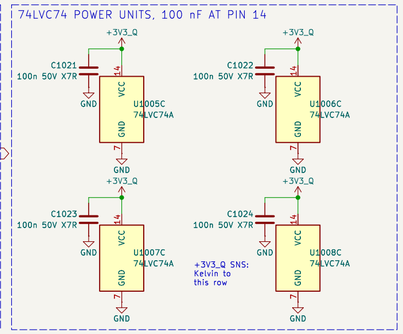
DC DETECTOR: OPENS BOTH RELAYS (F-mute-relay.md)



MCLK_Q BUFFER ON +3V3_Q (F-clk-buffer.md)

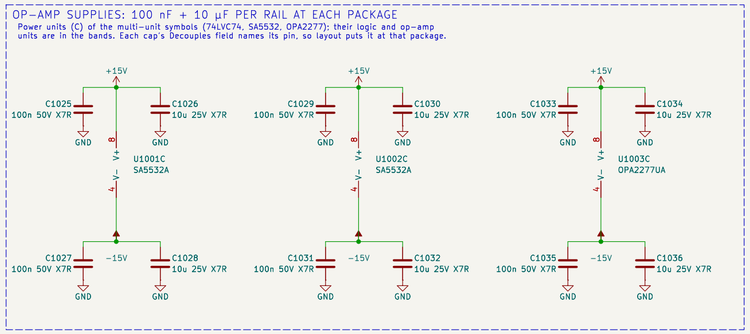


74LVC74 POWER UNITS, 100 nF AT PIN 14

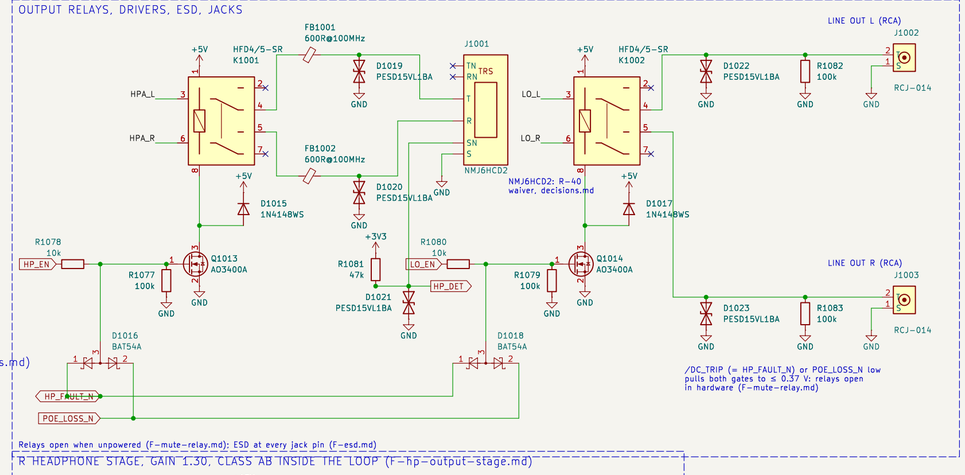


OP-AMP SUPPLIES: 100 nF + 10 µF PER RAIL AT EACH PACKAGE

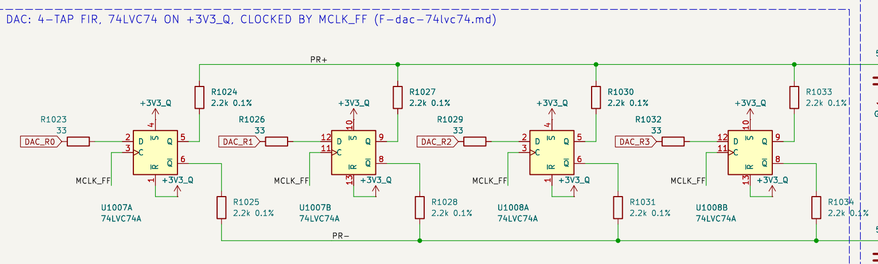
Power units (C) of the multi-unit symbols (74LVC74, SA5532, OPA2277): their logic and op-amp units are in the bands. Each cap's Decouples field names its pin, so layout puts it at that package.



OUTPUT RELAYS, DRIVERS, ESD, JACKS

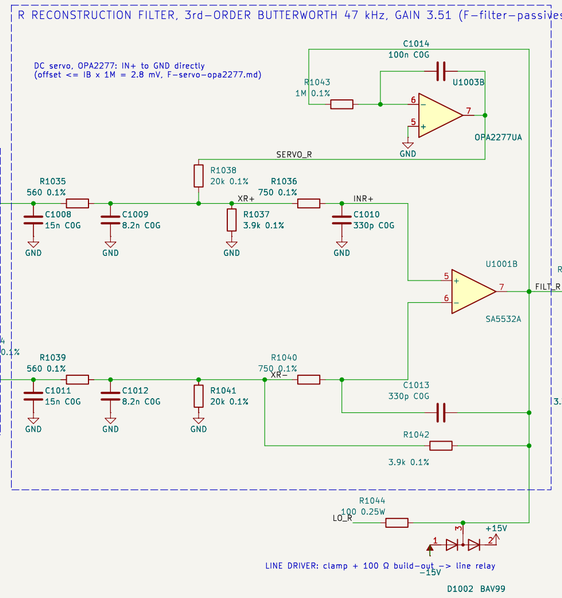


R 1-BIT DAC: 4-TAP FIR, 74LVC74 ON +3V3_Q, CLOCKED BY MCLK_FF (F-dac-74lvc74.md)

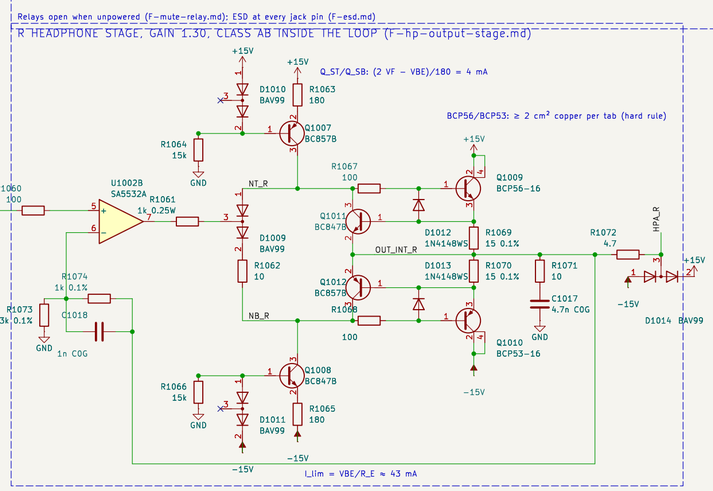


R_T 2.2k x 4 per side = 550 Ω; FIR zeros at 1.536 and 3.072 MHz

R RECONSTRUCTION FILTER, 3rd-ORDER BUTTERWORTH 47 kHz, GAIN 3.51 (F-filter-passives.md)



R HEADPHONE STAGE, GAIN 1.30, CLASS AB INSIDE THE LOOP (F-hp-output-stage.md)

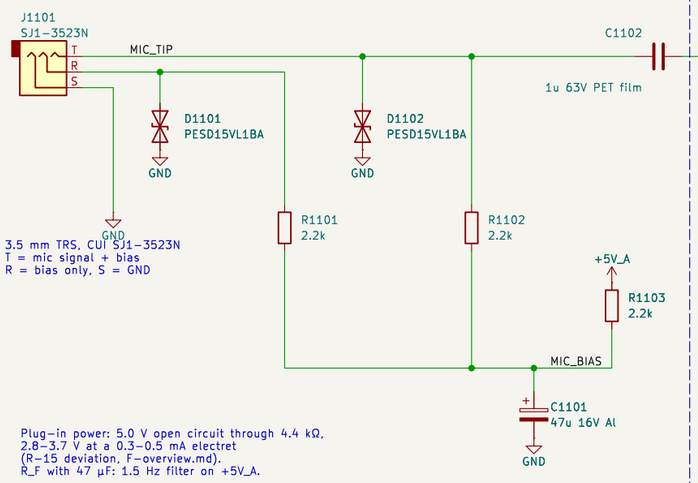


DAC/filter: F-dac-74lvc74.md, F-filter-passives.md; HP: F-hp-output-stage.md
Block F. Audits: design/parts/F-*.md
Sequoia Alexander, github.com/sequoia-hope/studio-amp
Sheet: /audio_out/
File: audio_out.kicad_sch

Title: Studio Amp satellite: 1-bit DAC, headphone and line out

Size: A2	Date: 2026-11-08	Rev: A
KiCad E.D.A. 9.0.8		Id: 12/14

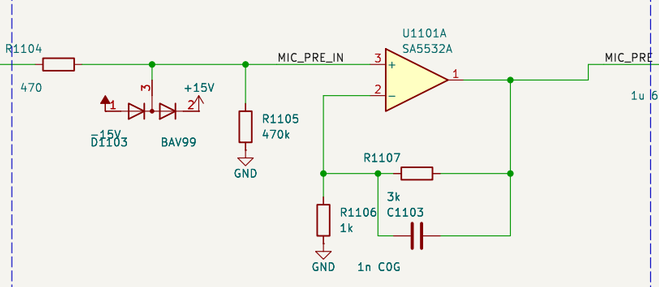
EXT MIC JACK, ESD AND PLUG-IN POWER (R-15)



3.5 mm TRS, CUI SJ1-3523N
T = mic signal + bias
R = bias only, S = GND

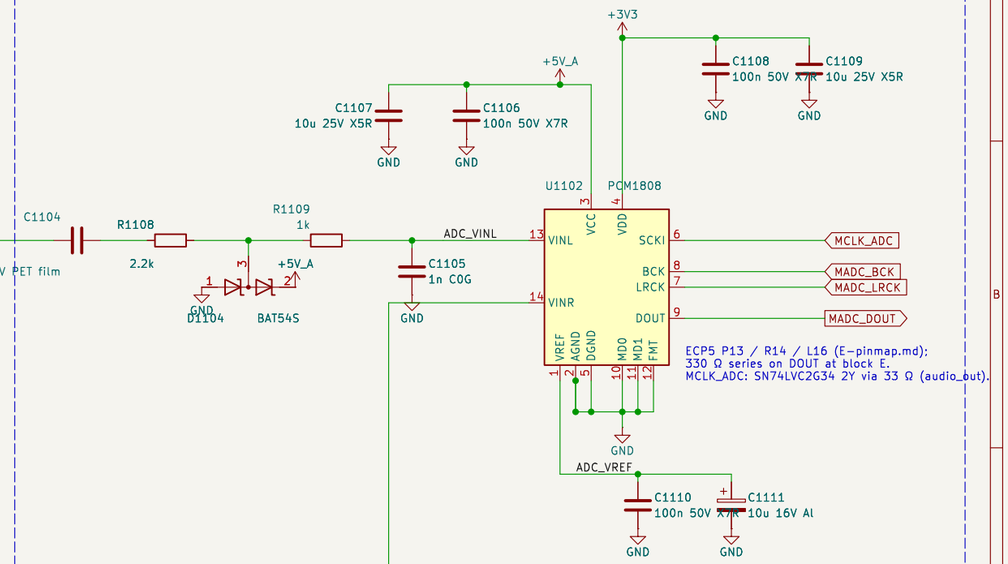
Plug-in power: 5.0 V open circuit through 4.4 kΩ,
2.8–3.7 V at a 0.3–0.5 mA electret
(R-15 deviation, F-overview.md).
R_F with 47 µF: 1.5 Hz filter on +5V_A.

MIC PREAMP, +12 dB (SA5532 A)



Gain 1 + 3k/1k = 4.0 (+12 dB): clips at 110 dB SPL
with a −28 dBV/Pa capsule. Line level plugged in
by mistake swings ±5.7 V, inside the input range.

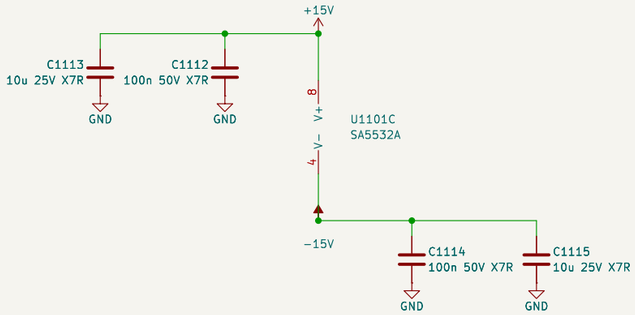
MIC ADC: PCM1808, I2S SLAVE, SCKI = 512 fs



ECP5 P13 / R14 / L16 (E-pinmap.md):
330 Ω series on DOUT at block E.
MCLK_ADC: SN74LVC2G34 2Y via 33 Ω (audio_out).

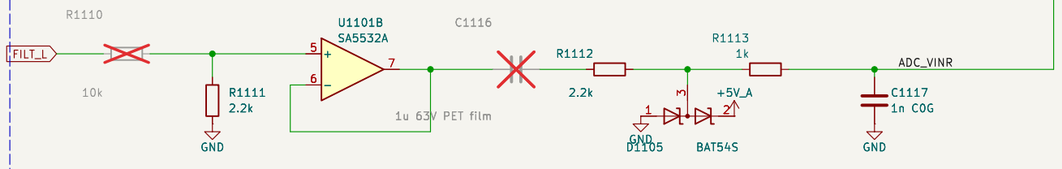
MD1 = MD0 = L: slave, SCKI auto-detect
FMT = L: I2S 24-bit

SA5532 SUPPLY (U1101)



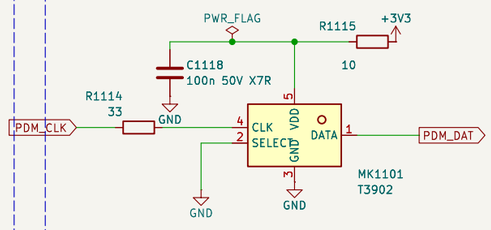
100 nF + 10 µF per rail within 3 mm of pins 8 and 4
(F-opamp-sa5532.md)

LOOPBACK: LINE-OUT L (FILT_L) -> PCM1808 VINR (OPTIONAL, R_LA AND C_LB DNP)



Not fitted: R_LA and C_LB. With them, VINR records line-out L
for latency and level calibration (room correction).

ON-BOARD MEMS MIC (R-16)



SELECT = GND: data on the right
channel (after CLK falls)

Bottom port: Ø0.8 mm PCB hole, vent to the back panel
(phase 4). No pull on DATA. Clock 3.072 MHz.

Mic chain: F-mic-adc-pcm1808.md; MEMS mic: F-mems-t3902.md
Block F. Audits: design/parts/F-*.md
Sequoia Alexander, github.com/sequoia-hope/studio-amp

Sheet: /audio_in/
File: audio_in.kicad_sch

Title: Studio Amp satellite: Measurement mic and MEMS mic

Size: A3	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 13/14

USB-C USB 2.0 FS to the RP2350, sink only (G-usb-c.md)

J1201
USB-C 16P TYPE-C-31-M-12

PWR_FLAG
TP1201 pad VUSB
VUSB
D1201 SMF5.0A
U1201 USBLC6-2SC6
R1205 15k
R1206 22
C1202 100n
VUSB_DET
USB_DM
USB_DP
27R series at the RP2350 (block E)

CC1 R1203 5.1k
CC2 R1204 5.1k

USB_SHIELD
R1201 0R
R1202 1M
C1201 4n7 2kV
Shell: 0R to GND fitted. Isolated option: remove the 0R, fit 1M || 4n7 2 kV (G-usb-c.md)

QWIIC TCA9617B repeater + TPS2553 switch, 155–207 mA (G-qwiic.md)

1 GND 2 +3V3 3 SDA 4 SCL (Qwiic)

J1202 Qwiic JST SH 4P
A-side 4.7k: drop if block E fits I2C1 pull-ups
TP1202 pad SCLB
TP1203 pad SDAB
R1207 2.2k
R1208 2.2k
C1203 100n
C1204 100n
R1209 4.7k
R1210 4.7k
I2C1_SCL
I2C1_SDA
QWIIC_PWR_EN
U1202 TCA9617B

JP1201 bridged
TP1204 pad QWIIC_3V3
U1203 TPS2553
C1205 1u 50V
C1206 10n 25V
R1213 100k
R1212 100k
R1214 10k
QWIIC_FAULT_N

U1204 SRV05-4
No series R on SDA/SCL: a far-end low plus I*R must stay below VILB 0.4 V

DEBUG PROBE JST SH 3-pin, Sequoia's convention (G-debug-jtag.md)

J1203 JST SH 3P
R1215 33R
D1202 PESD5V0S1BL
SWDIO
R1216 33R
D1203 PESD5V0S1BL
SWCLK
SWD: 1 SWCLK 2 GND 3 SWDIO

J1204 JST SH 3P
R1217 100R
D1204 PESD5V0S1BL
UART0_TX
R1218 100R
D1205 PESD5V0S1BL
UART0_RX
UART: 1 RX 2 GND 3 TX

ADDRESSABLE LED OUTPUT TPS2553 1.03–1.20 A, SN74LV1T125 (G-led-out.md)

RILIM 23.2k: 1.03–1.20 A; firmware caps it by PoE class
+5V_BUCK
U1205 TPS2553
C1207 1u 100V
+5V_LED
C1208 220u 25V
FB1201 BLM31KN601
D1206 SMF6.0A
TP1205 TP +5V_LED
+5V GND
J1209 JST PH 4P
R1220 232k
R1221 100k
LED_EN
LED_FAULT_N
R1219 100k

C1209 100n
LED_DATA
R1222 47R
TP1206 pad LED_DATA 5V
D1207 PESD5V0S1BL
U1206 SN74LV1T125

C1210 100n
LED_CLK
R1223 47R
TP1207 pad LED_CLK 5V
D1208 PESD5V0S1BL
U1207 SN74LV1T125

KNOB / ENCODER HEADER 2x6 0.1 in, optional (G-knob.md)

KNOB_REF
R1225 100R ADC_AVDD
KNOB_REF: RP2350 ADC_AVDD (block E) through 100R so knob readings are ratiometric. Pots 10–50k.
C1211 1u 50V
R1224 47R
J1206 2x6 header 2.54
KNOB_REF
KNOB1 R1226 1k D1209 PESD5V0S1BL C1212 100n
KNOB2 R1227 1k D1210 PESD5V0S1BL C1213 100n
KNOB3 R1228 1k D1211 PESD5V0S1BL C1214 100n
KNOB4 R1229 1k D1212 PESD5V0S1BL C1215 100n
ENC1_A R1230 1k D1213 PESD5V0S1BL R1231 10k C1216 10n
ENC1_B R1232 1k D1214 PESD5V0S1BL R1233 10k C1217 10n
ENC2_A R1234 1k D1215 PESD5V0S1BL R1235 10k C1218 10n
ENC2_B R1236 1k D1216 PESD5V0S1BL R1237 10k C1219 10n

FPGA JTAG 1x6 0.1 in, optional (G-debug-jtag.md)

1 +3V3 2 GND 3 TCK 4 TDO 5 TDI 6 TMS
4.7k pulls are on block E's sheet (TN-02038)
J1207 1x6 header 2.54
U1208 SRV05-4
JT_TMS R1238 33R FPGA_TMS
JT_TDI R1239 33R FPGA_TDI
JT_TDO R1240 33R FPGA_TDO
JT_TCK R1241 33R FPGA_TCK

DISPLAY HEADER 8-pin 0.1 in socket, every position remappable (G-display.md)

Each position: a jumper to DISP_VCC, DISP_BL_SW (switched backlight) or GND, else 47R to its FPGA pin. Defaults for GND, VCC, SCL, SDA, RES, DC, CS, BLK: JP to GND on 1, VCC on 2, BLK on 8 bridged. An FPGA line whose position is jumpered must be constrained HI-Z (block E).

J1208 1x8 socket 2.54
default: GND VCC SCL SDA RES DC CS BLK

	DISP_VCC	DISP_BL_SW		R 47R
DISP_P1	JP1202 open	JP1203 open	JP1204 bridged	R1242 → DISP_AUX7
DISP_P2	JP1205 bridged	JP1206 open	JP1207 open	R1243 → DISP_AUX8
DISP_P3	JP1208 open	JP1209 open	JP1210 open	R1244 → DISP_SCL
DISP_P4	JP1211 open	JP1212 open	JP1213 open	R1245 → DISP_SDA
DISP_P5	JP1214 open	JP1215 open	JP1216 open	R1246 → DISP_RES
DISP_P6	JP1217 open	JP1218 open	JP1219 open	R1247 → DISP_DC
DISP_P7	JP1220 open	JP1221 open	JP1222 open	R1248 → DISP_CS
DISP_P8	JP1223 open	JP1224 bridged	JP1225 open	R1249 → DISP_BLK

U1209 SRV05-4 (DISP_P1 DISP_P2 DISP_P3 DISP_P4)
U1210 SRV05-4 (DISP_P5 DISP_P6 DISP_P7 DISP_P8)

DISP_VCC AND BACKLIGHT SWITCH (G-display.md)

+3V3
FB1202 BLM18PG221
PWR_FLAG
DISP_VCC
TP1208 pad DISP_VCC
C1220 1u 50V
R1250 100k
Q1201 AO3401A
DISP_BL_SW
TP1209 pad DISP_BL_SW
PWM <= 2 kHz with the 100k gate pull-up (10k for faster PWM). Q off at power-up.
DISP_BL_PWM
R1251 100R
R1252 100k
Q1202 2N7002

TEST POINTS (R-80, G-testpoints.md) rail pads sit at their rails

LED_EN — TP1210 pad LED_EN — LED_FAULT_N — TP1211 pad LED_FAULT_N — QWIIC_PWR_EN — TP1212 pad QWIIC_PWR_EN — QWIIC_FAULT_N — TP1213 pad QWIIC_FAULT_N — VDRV_EN — TP1214 pad VDRV_EN — VDRV_PG — TP1215 pad VDRV_PG — PTP_PPS — TP1216 pad PTP_PPS — RUN — TP1217 pad RUN — TP1218 TP GND — TP1219 TP GND

Block G. Audits: design/parts/G-*.md
Sequoia Alexander, github.com/sequoia-hope/studio-amp
Sheet: /io/
File: io.kicad_sch

Title: Studio Amp satellite: User I/O, debug and test points

Size: A2	Date: 2026-11-08	Rev: A
KiCad E.D.A. 9.0.8		Id: 14/14